

Full-Stack Backend, Banking & Technical Glossary

A Plain-English Complete Guide covering Android Native, Node.js / Express on Vercel, Cryptographic Security, Banking Jargon & Legal Directives

Executive Full-Stack Overview: Loanzo operates a **Tri-Tier Hybrid Cloud Architecture**:

- 1. Client Tier:** Offline-first native Android app (Kotlin, Room SQLite, Jetpack Compose, WorkManager) with instantaneous zero-latency local commits.
- 2. Central Cloud Gateway Tier:** Serverless Node.js & Express.js v5 backend deployed on **Vercel** (backend-blond-sigma-66.vercel.app) acting as the secure gateway for DigiLocker KYC, Nodemailer SMTP OTPs, Meta WhatsApp Cloud webhooks, and Telegram Bot RBAC administration.
- 3. Distributed Cloud Datastore:** Google Cloud Firestore & Firebase Auth for cross-device state mirroring and Google Drive for encrypted KYC document vaults.

1. Core Software, Database & Architecture Terms

1. Entity [Database Structure]

What it means: A formal digital blueprint defining a single database record (one row in a SQLite table). In Loanzo, `UserEntity` holds user profile data, and `LoanEntity` stores loan contract details.

***Plain Analogy / Why it matters:** Think of an Excel spreadsheet named 'Users'. Each individual row containing a person's Name, Phone, and KYC status is an Entity.*

2. DAO (Data Access Object) [Database Layer]

What it means: The mediator interface (librarian) that safely queries or updates the SQLite database on behalf of the app screens without exposing raw SQL.

***Plain Analogy / Why it matters:** Like a waiter in a restaurant. You don't walk into the kitchen to prepare your food; you tell the waiter what you want, and the waiter fetches it safely.*

3. Offline-First / Local-First [System Paradigm]

What it means: An architectural standard where all operations write directly to the device's internal memory first before attempting network transit.

***Plain Analogy / Why it matters:** The app never freezes, lags, or displays loading spinners in basements or rural areas with poor connectivity. Sync happens quietly in the background when signal returns.*

4. Reactive Flow (Flow) [State Pipeline]

What it means: A live asynchronous stream that automatically emits updated data to UI components the instant a change occurs in the underlying SQLite database.

***Plain Analogy / Why it matters:** Like a live sports scoreboard or stock ticker. You never have to press 'Refresh' (F5); the moment an EMI is logged in the database, the screen instantly reflects it.*

5. Idempotent Operations [Distributed Systems]

What it means: A mathematical guarantee that an action produces the identical result whether executed once or 100 times.

***Plain Analogy / Why it matters:** Pressing an elevator button that is already lit stays on floor 4. If poor cellular reception causes a payment sync request to transmit twice, the backend ensures the user is charged only once.*

6. WorkManager & SyncWorker [Android OS Services]

What it means: The resilient Android OS background scheduler that guarantees execution of tasks even if the app process is terminated or the phone reboots.

***Plain Analogy / Why it matters:** Android's personal courier. It holds pending offline transactions in a queue and wakes up automatically when internet is detected to complete the sync.*

2. Central Node.js, Express.js & Vercel Backend Subsystems

Deployed at: <https://backend-blond-sigma-66.vercel.app> | Local Source: C:\Users\hp\OneDrive\Desktop\Loanzo\backend

7. Vercel Serverless Hosting (@vercel/node) [Cloud Infrastructure]

What it means: The serverless cloud platform hosting Loanzo's central backend. Configured via vercel.json to route all inbound traffic /(.*) into server.js.

***Plain Analogy / Why it matters:** Zero-server maintenance. Instead of paying for an idle virtual machine, Vercel spins up lightweight execution workers on demand with global edge CDN caching and automated SSL.*

8. Express.js v5 API Engine (server.js) [Backend Framework]

What it means: A high-performance Node.js web server handling 1,686 lines of production routing, middleware, CORS security, request validation, and external service orchestration.

***Plain Analogy / Why it matters:** The central traffic controller. It acts as the bridge connecting Android app requests with government APIs, email servers, Telegram channels, and WhatsApp webhooks.*

9. DigiLocker & Sandbox.co.in Regulatory Proxy [KYC Gateway]

What it means: A secure backend proxy (/api/kyc/digilocker/*) that initiates KYC sessions, requests user consent, and retrieves real UIDAI Aadhaar XML and Income Tax PAN cards.

***Plain Analogy / Why it matters:** Secret Key Vault. By isolating Sandbox.co.in API secret keys on the Vercel backend rather than compiling them into the Android APK, hackers cannot decompile the app to steal government credentials.*

10. Nodemailer SMTP OTP Dispatch Engine [Email Authentication]

What it means: A server-side service (/api/auth/send-email-otp) integrating with Gmail SMTP to generate cryptographically random 6-digit verification codes and format institutional HTML emails.

***Plain Analogy / Why it matters:** When a user signs up on the Android app, this endpoint delivers the 6-digit confirmation code directly to their inbox within seconds.*

11. Interactive Telegram RBAC Command Center [Operations & Telemetry]

What it means: An 840+ line interactive Telegram bot (/api/telegram/webhook & /notify) providing real-time loan approval alerts, dispute alerts, and inline buttons ([Approve], [Reject], [Freeze]).

***Plain Analogy / Why it matters:** A mobile command center for the App Owner. Admins can audit high-value transactions and trigger emergency account freezes right from Telegram without opening the app.*

12. Meta WhatsApp Cloud API Webhooks [Borrower Messaging]

What it means: Inbound and outbound webhook handlers (/webhook & /api/webhook) connecting directly to Meta's WhatsApp Business Platform.

***Plain Analogy / Why it matters:** Sends automated official WhatsApp loan updates, EMI payment reminders, and legally signed agreement download links directly to borrowers' chat windows.*

3. Banking, Credit & Financial Mechanics

13. Amortization & Reducing Balance EMI [Lending Math]

What it means: The mathematical calculation schedule where monthly EMI payments are split between paying down accrued interest and reducing the principal borrowed.

***Plain Analogy / Why it matters:** As you repay, your principal balance shrinks. Next month's interest is calculated ONLY on the remaining smaller balance, lowering the borrower's total interest cost over time.*

14. LTV (Loan-to-Value) Ratio [Risk Management]

What it means: The regulatory ceiling governing the maximum loan amount that may be extended against the appraised market value of pledged collateral.

***Plain Analogy / Why it matters:** If gold jewelry is valued at ₹1,00,000, RBI regulations mandate a 75% LTV cap, meaning the maximum loan permitted is ₹75,000. The 25% safety margin protects the lender against gold price fluctuations.*

15. Bank Account 'Penny-Drop' Verification [Fraud Prevention]

What it means: An automated micro-deposit of ₹1.00 dispatched via IMPS to the borrower's receiving bank account before loan approval.

***Plain Analogy / Why it matters:** The receiving bank automatically responds with the legal registered name of the account holder. If this name does not match the borrower's government Aadhaar/PAN, the transfer is aborted to prevent mule accounts.*

16. Casual Lending Guidelines (RBI Directives) [Regulatory Law]

What it means: Statutory caps established by the Reserve Bank of India to govern informal peer-to-peer credit and prevent unregulated loan-sharking.

***Plain Analogy / Why it matters:** Enforces maximum total exposure limits on non-institutional lenders, ensuring consumer protection and stopping usurious compounding interest.*

4. Security, Hardware & Cryptographic Subsystems

17. SHA-256 Cryptographic Hash [Data Integrity]

What it means: A one-way mathematical function that transforms any document or data set into a unique, fixed 64-character digital fingerprint.

***Plain Analogy / Why it matters:** Like an unforgeable wax seal. If anyone alters a single punctuation mark or decimal point in a signed 30-page loan contract, the fingerprint changes completely, proving tampering in court.*

18. Cryptographic Hash Chain (Audit Trail) [Tamper-Proof Ledger]

What it means: A chained ledger structure where each sequential transaction references and incorporates the mathematical hash of the preceding event.

***Plain Analogy / Why it matters:** A personal mini-blockchain. Event 1 (Disbursement) links to Event 2 (Repayment). An administrator or hacker cannot delete or edit Event 1 without invalidating the entire subsequent chain.*

19. Hardware Device Fingerprint & Binding [Device Attestation]

What it means: Extracting unique physical smartphone hardware attributes (IMEI, build serial, secure enclave ID) to bind the banking session to that specific phone.

***Plain Analogy / Why it matters:** Even if a cybercriminal steals your login ID and password, they cannot log into your account from a laptop or secondary phone without possessing your physical registered device.*

20. Biometric CryptoObject [Hardware Security]

What it means: An Android security architecture where fingerprint or facial biometric authentication unlocks a cryptographic key inside the phone's hardware security chip (Titan M / Knox).

***Plain Analogy / Why it matters:** The biometrics do not just provide a simple Yes/No check; they physically release an encrypted key from silicon hardware to authorize transactions.*

5. Legal Frameworks, Physical Custody & Operations

21. Leegality & Aadhaar eSign [Digital Contracts]

What it means: A legal digital signing rail backed by UIDAI OTP verification on the signer's Aadhaar-linked mobile phone.

***Plain Analogy / Why it matters:** Under Section 5 of the Information Technology Act (2000), an Aadhaar eSign carries the exact same judicial enforceability as an ink signature on notarized stamp paper.*

22. Section 65B Certificate (Indian Evidence Act) [Judicial Admissibility]

What it means: A statutory certificate required by Indian courts to admit electronic records (app logs, PDF dossiers, timestamps) as valid primary evidence in legal recovery proceedings.

***Plain Analogy / Why it matters:** Loanzo's LegalDossierExportEngine automatically embeds device IDs, UTC hashes, and server metadata so an unpaid default can be presented directly to an arbitrator or judge without being dismissed as raw text.*

23. TESB (Tamper-Evident Security Bag) [Physical Custody]

What it means: A high-security, serialized custody pouch used by gold loan companies and vault operators featuring irreversible 'VOID' tamper indicators.

***Plain Analogy / Why it matters:** Once sealed, any attempt to peel, slice, or heat the bag permanently displays a visible 'VOID / OPENED' watermark. Both borrower and field agent must scan its unique barcode on-site.*

24. Out-of-Band (OOB) Verification & Silent Duress [Anti-Extortion]

What it means: Communicating with a party over a separate, private communication channel that a third party standing nearby cannot monitor or manipulate.

***Plain Analogy / Why it matters:** If a corrupt field agent attempts to extort cash from a borrower at their home, the platform alerts the borrower directly on their personal device, allowing them to secretly report extortion without escalating danger.*

25. Joint & Several Liability (Section 128 ICA & IPC 120B) [Syndicate Defense]

What it means: A legal doctrine making both co-conspirators individually and collectively responsible for the entire financial loss and criminal penalties.

Plain Analogy / Why it matters: If an agent and borrower conspire to submit fake collateral, the agent is legally classified as an attesting co-guarantor. The lender can recover funds directly from the agent's account, and both face criminal conspiracy prosecution.

6. Architectural Subsystem Mapping Table (Android & Vercel)

Subsystem	Primary Code Files	Operational Responsibility
Vercel Backend API	server.js, package.json, vercel.json	Express.js gateway on Vercel: DigiLocker KYC proxy, email OTP, Tele
Local Database	LoanzoDatabase.kt, 15 DAOs, 21 Entities	Zero-latency offline SQLite storage with continuous reactive Flow emis
Sync Engine	AppSyncManager.kt, SyncWorker.kt	Background offline-to-cloud queueing with WorkManager & Firestore s
Identity & KYC	DigiLockerConfig.kt, DiditService.kt	Calls Vercel /api/kyc/digilocker for government Aadhaar XML & AI liver
Contracts & Sign	LeegalityService.kt, LoanzoPdfEngine.kt	Aadhaar eSign and court-admissible Sec 65B legal PDF dossiers.
Domain Rules	RuleEngine.kt, CasualLendingGuard.kt	RBI casual lending limits, reducing balance amortization, and risk chec
Banking Session	BankingSessionManager.kt, DeviceSecurityHelper.kt	Hardware IMEI binding, 3-min auto-lock, and 48-hr hard session expiry